

Optimal Sampling and Actuation Policies of a Markov Source over a Wireless Channel

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Abstract—This paper studies efficient data management and timely information dissemination for real-time monitoring of an N -state Markov process, enabling accurate state estimation and reliable actuation decisions. First, we analyze the Age of Incorrect Information (AoII) and derive closed-form expressions for its time average under several scheduling policies, including randomized stationary, change-aware randomized stationary, semantics-aware randomized stationary, and threshold-aware randomized stationary policies. We then formulate and solve constrained optimization problems to minimize the average AoII under a time-averaged sampling action constraint, and compare the resulting optimal sampling and transmission policies to identify the conditions under which each policy is most effective. We further show that directly using reconstructed states for actuation can degrade system performance, especially when the receiver is uncertain about the state estimate or when actuation is costly. To address this issue, we introduce a cost function, termed the *Cost of Actions under Uncertainty* (CoAU), which determines when the actuator should take correct actions and avoid incorrect ones when the receiver is uncertain about the reconstructed source state. We propose a *randomized actuation policy* and derive a closed-form expression for the probability of taking no incorrect action. Finally, we formulate an optimization problem to find the optimal randomized actuation policy that maximizes this probability. The results show that the resulting policy substantially reduces incorrect actuator actions.

I. INTRODUCTION

Time-critical networked systems rely on continuous sensing of physical processes to support applications such as health-care monitoring, autonomous transportation, and industrial automation [1]–[4]. The effectiveness of these applications depends not only on receiving updates but also on receiving information that is *fresh*, *relevant*, and *informative* for timely decision-making or actuation. As these systems generate large volumes of data, transmitting every update without considering its relevance to the underlying task or its impact on decision-making is inefficient and unsustainable. Such indiscriminate communication leads to unnecessary energy and bandwidth consumption and can degrade overall system performance. Therefore, efficient data management that prioritizes task-relevant information is essential for optimal operation. To

address this challenge, there has been a shift from conventional data-centric transmission policies toward *goal-oriented*, *semantics-aware communication*. This paradigm leverages semantics-aware metrics to jointly design *data generation*, *transmission*, and *utilization strategies*, enabling timely and informative updates under limited transmission resources to achieve task-specific objectives [5]–[9].

The Age of Information (AoI) is a semantics-aware metric that quantifies the timeliness, or freshness, of received information [10]; it is defined as the time elapsed since the generation of the most recently received update. However, AoI captures only freshness and does not account for the information’s content or significance. To address this limitation, the Age of Incorrect Information (AoII) was introduced in [11] to measure the duration for which the system operates with an incorrect state estimate, thereby capturing both timeliness and inaccuracy. Beyond AoI and AoII, several other semantics-aware metrics have been proposed [12]–[25] to evaluate information quality in terms of timeliness, relevance, and significance. The primary goal of these studies is to design sampling and transmission strategies that optimize semantics-aware metrics under various system constraints, demonstrating substantial improvements in the efficiency and effectiveness of real-time monitoring. In these works, decisions at the actuator are typically based on the reconstructed state at the receiver or on the received updates. However, even when semantics-aware metrics are optimized, directly acting on the reconstructed state can lead to severe performance degradation, particularly when the receiver is uncertain about the correctness of the reconstructed source state or when actuation is costly. To overcome this limitation, we propose a cost function, termed *Cost of Actions under Uncertainty* (CoAU), which ensures the actuator performs correct actions while preventing incorrect actions when the receiver is uncertain about the correctness of the reconstructed source state.

In this work, we study the problem of real-time remote monitoring of an N -state Markov process, where a sampler observes the state of the process and decides whether to sample. We consider a set of sampling policies, namely randomized stationary, change-aware randomized stationary, semantics-aware randomized stationary, and threshold-aware randomized stationary policies. When sampling is performed, the trans-

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mitter sends the sample as a packet to the receiver over an unreliable wireless channel; otherwise, the transmitter remains idle. The receiver then performs real-time state estimation based on the successfully received samples, and the actuator decides whether to take an action based on the reconstructed state at the receiver. To evaluate system performance, we derive closed-form expressions for the average AoII under the proposed joint sampling and transmission policies. For actuation decisions, we propose a *randomized actuation* policy and derive a closed-form expression for the probability that the CoAU equals zero under this policy. Furthermore, we formulate and solve two constrained optimization problems. The first problem aims to minimize the average AoII under a constraint on the time-averaged sampling action. In the second problem, the objective is to maximize the probability that the CoAU equals zero, subject to a constraint on the time-averaged actuation action.

II. SYSTEM MODEL

We consider a remote monitoring system that includes an information source, a sampler, a transmitter, a receiver, and an actuator, as shown in Fig. 1. The system operates over discrete time, which we represent as a sequence of equally spaced time slots indexed by $t \in \mathbb{N}$. The state of the information source at time slot t is denoted by $X(t)$, and it is modeled as an N -state discrete-time Markov chain (DTMC). This state represents information that can be mapped to an action at the receiver side, or it can be considered as a control set that determines the actuation at the actuation point. Specifically, the self-transition probability and the probability of transitioning to a different state at time slot $t + 1$ are defined as $\mathbb{P}[X(t + 1) = i | X(t) = i] = q$ and $\mathbb{P}[X(t + 1) = j | X(t) = i] = p, \forall i, j \in \{0, 1, \dots, N - 1\}$ with $i \neq j$, respectively, where $q + (N - 1)p = 1$.

At the beginning of each time slot t , the sampler observes the current state of the information source and decides whether to perform sampling. The sampling action at time slot t is represented by $\alpha(t) \in \{0, 1\}$, where $\alpha(t) = 1$ indicates that the source is sampled and $\alpha(t) = 0$ indicates that no sampling occurs. When the source is sampled, the transmitter immediately sends the sample as a packet over the wireless communication channel to the receiver; otherwise, the transmitter remains idle. We assume that the transmission is successful when the channel state $h(t)$ equals 1, while $h(t) = 0$ indicates a decoding failure. The probability of successful decoding is defined as $p_s = \mathbb{P}[h(t) = 1]$. At the end of time slot t , the receiver constructs an estimate of the information source state, denoted by $\hat{X}(t)$, based on the successfully received samples. If the receiver successfully decodes the transmitted sample, then $X(t) = \hat{X}(t)$; otherwise, the reconstructed state remains unchanged, i.e., $\hat{X}(t) = \hat{X}(t - 1)$. Likewise, if the transmitter does not send a sample or if the transmission fails, it is assumed that $\hat{X}(t) = \hat{X}(t - 1)$. At time slot t , the system is considered to be in a synchronized (synced) state if $X(t) = \hat{X}(t)$, and in an erroneous state if $X(t) \neq \hat{X}(t)$. Acknowledgment (ACK) and negative-acknowledgment

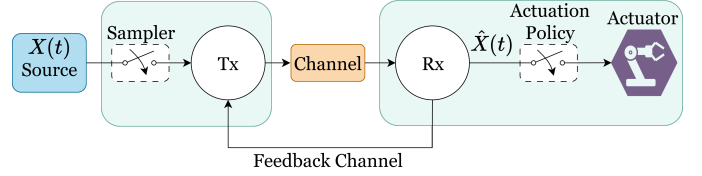


Fig. 1. Real-time monitoring of a Markovian source over a wireless channel.

(NACK) packets are used to inform the transmitter about the success or failure of transmissions, and these feedback packets are assumed to be delivered instantly and without errors. Therefore, the transmitter has perfect knowledge of the receiver's reconstructed state at each time slot. In addition, we assume that any sample from an unsuccessful transmission is discarded. The actuator then decides whether to perform an action based on the reconstructed source state. The actuator's action at time slot t is denoted by $c(t) \in \{0, 1\}$, where $c(t) = 1$ indicates that the actuator performs an action, and $c(t) = 0$ indicates that it remains idle.

III. PERFORMANCE METRICS AND ANALYSIS

In this section, we investigate the impact of incorrect information and incorrect actions at the receiver and actuator, respectively. We first consider the *Age of Incorrect Information* (AoII), which captures both timeliness, inaccuracy, and significance of information at the receiver. However, making actuation decisions directly based on the reconstructed state may pose safety risks, particularly when the receiver is uncertain about the correctness of the reconstructed source state or when actuation incurs high costs. To address this, we introduce the *Cost of Actions under Uncertainty* (CoAU), which explicitly quantifies incorrect actuation decisions.

A. Age of Incorrect Information (AoII)

The AoII measures the time that has passed since the transmitted information became incorrect or outdated [11]. Let $\text{AoII}(t)$ denote the AoII at time slot t . When the system is in a synced state at time t , $\text{AoII}(t) = 0$; otherwise, $\text{AoII}(t) = i \geq 1$ indicates that the system has been in an erroneous state from time slot $t - i + 1$ to t . The evolution of AoII is defined as follows:

$$\text{AoII}(t) = \begin{cases} \text{AoII}(t - 1) + 1, & X(t) \neq \hat{X}(t), \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

To evaluate the performance of the AoII metric, we consider the following joint sampling and transmission policies:

- 1) **Threshold-aware randomized stationary (TARS):** In this policy, at time slot t , if the AoII is greater than or equal to a threshold n , the sampler always performs sampling. If the AoII equals $n - 1$, the sampler performs sampling with probability p_α^{th} ; otherwise, the sampler remains idle.
- 2) **Randomized Stationary (RS):** In this policy, a new sample is generated at each time slot t with probability p_α^r , regardless of whether the system is synced.
- 3) **Change-aware randomized stationary (CARS):** In this policy, at time slot t , no sample is taken if the source state has not changed, i.e., $X(t) = X(t - 1)$. Otherwise,

if $X(t) \neq X(t-1)$, a new sample is generated with probability p_α^c .

- 4) Semantics-aware randomized stationary (SARS): In this policy, at each time slot, the generation of a new sample is triggered probabilistically only when the system is in an incorrect state, i.e., $X(t) \neq \hat{X}(t-1)$. Specifically, if $X(t-1) = \hat{X}(t-1)$ and $X(t) \neq X(t-1)$, the sampler generates a sample at time slot t with probability q_{α_1} . Furthermore, if $X(t-1) \neq \hat{X}(t-1)$ and $X(t) \neq \hat{X}(t-1)$, the sampler generates a sample at time slot t with probability q_{α_2} , where $q_{\alpha_2} \geq q_{\alpha_1}$.

Lemma 1. For an N -state DTMC information source, the average AoII under the threshold-aware randomized stationary policy is given by:

$$\overline{\text{AoII}} = \frac{F(p_\alpha^{th}, n)}{G(p_\alpha^{th}, n)}, \quad (2)$$

where $F(p_\alpha^{th}, n)$ and $G(p_\alpha^{th}, n)$ are given by:

$$\begin{aligned} F(p_\alpha^{th}, n) &= (N-1) \left[(1-p)(p+(1-p)p_s)^2 + (1-p)^n p_s \right. \\ &\quad \times \left(3p^2 - 2p - np^2 - p^3 + np^3(1-p_\alpha^{th}) - p^2 p_\alpha^{th} + p^3 p_\alpha^{th} \right. \\ &\quad \left. \left. + (1-p)p_s(1+(n-2)p - (n-1)p^2(1-p_\alpha^{th})) \right) \right], \\ G(p_\alpha^{th}, n) &= p(p_s + (1-p_s)p) [N(1-p)(p_s + (1-p_s)p) \\ &\quad - (N-1)(1-p)^n (1 - (1-p_\alpha^{th})p)p_s]. \end{aligned} \quad (3)$$

Furthermore, under the RS policy, the average AoII can be expressed as follows:

$$\overline{\text{AoII}} = \frac{(N-1)p(1-p_\alpha^r p_s)}{[p + (1-p)p_\alpha^r p_s][p_\alpha^r p_s + Np(1-p_\alpha^r p_s)]}. \quad (4)$$

Moreover, the average AoII for the SARS policy can be calculated as follows:

$$\overline{\text{AoII}} = \frac{(N-1)p(1-q_{\alpha_1} p_s)}{[p + (1-p)q_{\alpha_2} p_s][Np + (1-N)pq_{\alpha_1} p_s + (1-p)q_{\alpha_2} p_s]}. \quad (5)$$

For the CARS policy, the average AoII is obtained as follows:

$$\overline{\text{AoII}} = \frac{(N-1)(1-p_\alpha^c p_s)}{p(N-p_\alpha^c p_s)[1 + (N-2)p_\alpha^c p_s]}. \quad (6)$$

Proof: See Appendix A. ■

B. Cost of Actions under Uncertainty (CoAU)

The cost of actions under uncertainty, arising from the receiver's inability to determine whether the reconstructed state accurately reflects the true source state, is a critical challenge in real-time remote monitoring systems, especially in the presence of high error rates or costly actuation actions. We have presented several joint sampling and transmission policies that efficiently utilize sampling and transmission resources to reduce system error. To further improve system performance, we now focus on the cost of actions under uncertainty to prevent the actuator from making incorrect decisions based on the reconstructed state at the receiver. To this end, we propose

a *randomized actuation* policy in which the actuator takes actions *probabilistically* at each time slot t : if the receiver successfully decodes the transmitted information, an action is taken with probability p_{c_1} ; otherwise, an action is taken with probability p_{c_2} . In this context, the system may be synced while the receiver is unable to correctly infer that the reconstructed state matches the true state of the source. This mismatch can arise when the system is in a synced state, but the sampling and transmission policy either prevents the transmitter from sending an update or the transmitted update fails due to channel errors, leaving the receiver uncertain about the system's synced state. Therefore, an appropriate selection of the actuation probabilities p_{c_1} and p_{c_2} is essential to prevent incorrect actions. To capture the number of consecutive time slots during which the actuator performs incorrect actions, we define a cost function called the *Cost of Actions under Uncertainty* (CoAU). The objective of this cost function is to ensure that the actuator takes correct actions while preventing incorrect actions when the receiver is uncertain about the correctness of the estimated state of the source. Let $\Delta(t)$ denote the CoAU at time slot t . If $\Delta(t) = 0$, the actuator performs a correct action at time slot t , whereas $\Delta(t) = i \geq 1$ indicates that from time slot $t-i+1$ to t , the actuator has performed incorrect actions consecutively. According to this metric, $\Delta(t)$ increases by one when the system is synced, i.e., $X(t) = \hat{X}(t)$, but the actuator does not perform an action, $c(t) = 0$, or when the system is in an erroneous state, i.e., $X(t) \neq \hat{X}(t)$, but the actuator performs an action, $c(t) = 1$. In all other cases, $\Delta(t)$ drops to 0. We describe the evolution of the CoAU as follows:

$$\Delta(t) = \begin{cases} 0, & \begin{aligned} &\{\alpha(t)=1, h(t)=1, X(t)=\hat{X}(t), c(t)=1\}, \\ &\text{or} \{\alpha(t)=1, h(t)=0, X(t)=\hat{X}(t), c(t)=1\}, \\ &\text{or} \{\alpha(t)=0, X(t)=\hat{X}(t), c(t)=1\}, \\ &\text{or} \{X(t) \neq \hat{X}(t), c(t)=0\}, \end{aligned} \\ \Delta(t-1)+1, & \begin{aligned} &\{X(t)=\hat{X}(t), c(t)=0\}, \\ &\text{or} \{X(t) \neq \hat{X}(t), c(t)=1\}. \end{aligned} \end{cases} \quad (7)$$

Lemma 2. When the information source is sampled using the RS policy, the probability $P_{\Delta_0}^{RS} = \mathbb{P}[\Delta(t)=0]$ is given by:

$$\begin{aligned} P_{\Delta_0}^{RS} &= \frac{(N-1)p(1-p_\alpha^r p_s)}{Np + (1-Np)p_\alpha^r p_s} + p_\alpha^r p_s p_{c_1} \\ &\quad + \frac{(1-p_\alpha^r p_s)[2p - Np + (1-Np)p_\alpha^r p_s]p_{c_2}}{Np + (1-Np)p_\alpha^r p_s}. \end{aligned} \quad (8)$$

Proof: See Appendix B. ■

Similar to Lemma 2, one can obtain the probability $\mathbb{P}[\Delta(t)=0]$ when the source is sampled under the SARS and CARS policies as follows:

$$\begin{aligned}
P_{\Delta_0}^{\text{SARS}} &= \frac{(N-1)p(1-q_{\alpha_1}p_s)}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \\
&+ \frac{(N-1)pp_s[p(q_{\alpha_1}-q_{\alpha_2})+q_{\alpha_2}]p_{c_1}}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \\
&+ \frac{p[2-N+(N-1)(1-p)q_{\alpha_1}p_s]p_{c_2}}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \\
&+ \frac{(1-p)(1+(1-N)p)q_{\alpha_2}p_s p_{c_2}}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s}, \quad (9)
\end{aligned}$$

$$\begin{aligned}
P_{\Delta_0}^{\text{CARS}} &= \frac{(N-1)(1-p_{\alpha}^c p_s)}{N-p_{\alpha}^c p_s} + (N-1)pp_{\alpha}^c p_s p_{c_1} \\
&+ \left[3-2N-(N-1)pp_{\alpha}^c p_s + \frac{2(N-1)^2}{N-p_{\alpha}^c p_s} \right] p_{c_2}. \quad (10)
\end{aligned}$$

IV. CONSTRAINED OPTIMIZATION PROBLEMS

In this section, we formulate and solve two constrained optimization problems. In the first problem, presented in Section IV-A, the objective is to find the optimal TARS, RS, SARS, and CARS sampling policies that minimize the average AoII, while ensuring that the time-averaged sampling action remains below a given threshold. The second problem, presented in Section IV-B, aims to determine the optimal actuation probabilities that maximize the probability of correct actions at the actuator, subject to a constraint on the time-averaged actuation action.

A. Minimizing the average AoII

The objective of this optimization problem is to determine the optimal TARS sampling policy and the corresponding sampling probabilities p_{α}^r , p_{α}^c , q_{α_1} , and q_{α_2} for the RS, SARS, and CARS policies, respectively. The aim is to minimize the average AoII while satisfying a constraint on the time-averaged sampling action. Here, the parameter $0 < \eta \leq 1$ denotes the prescribed threshold for the time-averaged sampling action. First, focusing on the TARS sampling policy, we formulate the optimization problem as follows:

$$\text{minimize}_{p_{\alpha}^{th}, n} \frac{F(p_{\alpha}^{th}, n)}{G(p_{\alpha}^{th}, n)} \quad (12a)$$

$$\text{subject to } \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \leq \eta, \quad (12b)$$

where $\mathbf{1}$ denotes the indicator function, and $F(p_{\alpha}^{th}, n)$ and $G(p_{\alpha}^{th}, n)$ are given in (3). For an N -state DTMC information source, the time-averaged sampling action under the TARS policy can be expressed as:

$$\begin{aligned}
&\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \\
&= \begin{cases} \frac{p(N-1) + (p-(Np-1)p_s)p_{\alpha}^{th}}{Np + (1-p)p_s - (N-1)pp_{\alpha}^{th}p_s}, & n = 1, \\ \frac{(N-1)p(1-p)^{n-1}(1-(1-p_{\alpha}^{th})p)}{N(1-p)(p+(1-p)p_s) - (N-1)(1-p)^n(1-(1-p_{\alpha}^{th})p)p_s}, & n \geq 2. \end{cases} \quad (13)
\end{aligned}$$

The optimal value of n , denoted by n^* , is the minimum value of n that satisfies the constraint in (12b). To determine n^* , we first assume $p_{\alpha}^{th} = 0$ and find the minimum value of n that satisfies this constraint. Then, by substituting this optimal n into (12b), we obtain the optimal sampling probability that satisfies the optimization constraint. Using (13), if $\frac{p(N-1)}{Np + (1-p)p_s} \leq \eta$, the optimal value of n is $n^* = 1$ and the optimal sampling probability is $p_{\alpha}^{th*} = \min \left\{ 1, \frac{\eta(Np + (1-p)p_s) - p(N-1)}{p + \eta(N-1)pp_s - (Np-1)p_s} \right\}$. Otherwise, the optimal values of n and p_{α}^{th} that satisfy the constraint of the optimization problem are given as follows:

$$\begin{aligned}
n^* &= \left\lceil \frac{\log \left(\frac{N(1-p)(p+(1-p)p_s)\eta}{(N-1)p + (N-1)(1-p)p_s\eta} \right)}{\log(1-p)} \right\rceil, \\
p_{\alpha}^{th*} &= \max \left\{ 0, \frac{1-p}{p} \left[\frac{\eta N(1-p)(p_s + (1-p_s)p)}{(N-1)(1-p)^{n^*}(p + (1-p)\eta p_s)} - 1 \right] \right\}. \quad (14)
\end{aligned}$$

Now, using (4), the optimization problem for the RS policy is formulated as follows:

$$\text{minimize}_{p_{\alpha}^r} \frac{(N-1)p(1-p_{\alpha}^r p_s)}{[p + (1-p)p_{\alpha}^r p_s][p_{\alpha}^r p_s + Np(1-p_{\alpha}^r p_s)]} \quad (15a)$$

$$\text{subject to } \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \leq \eta, \quad (15b)$$

where the constraint in (15b) represents the time-averaged sampling action, which can be simplified as:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] = p_{\alpha}^r. \quad (16)$$

Using (16), the optimization problem in (15) is simplified as:

$$\text{minimize}_{p_{\alpha}^r} \frac{(N-1)p(1-p_{\alpha}^r p_s)}{[p + (1-p)p_{\alpha}^r p_s][p_{\alpha}^r p_s + Np(1-p_{\alpha}^r p_s)]} \quad (17a)$$

$$\text{subject to } p_{\alpha}^r \leq \eta. \quad (17b)$$

To solve this optimization problem, we note that the objective function in (17a) decreases as p_{α}^r increases. Therefore, the problem reduces to finding the maximum value of p_{α}^r that satisfies the constraint in (17b). Using this constraint, the optimal sampling probability is η . Furthermore, using (5), we can formulate the corresponding optimization problem for the SARS policy as follows:

$$\text{minimize}_{q_{\alpha_1}, q_{\alpha_2}} \frac{(N-1)p(1-q_{\alpha_1}p_s)}{[p+(1-p)q_{\alpha_2}p_s][Np+(1-N)pq_{\alpha_1}p_s+(1-p)q_{\alpha_2}p_s]} \quad (18a)$$

$$\text{subject to } \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \leq \eta. \quad (18b)$$

For an N -state DTMC information source, the time-averaged sampling action under the SARS policy can be expressed as:

$$\begin{aligned} & \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \\ &= \frac{p(N-1)[p(q_{\alpha_1} - q_{\alpha_2}) + q_{\alpha_2}]}{Np - (N-1)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s}. \end{aligned} \quad (19)$$

Using (19), the optimization problem in (18) is simplified as:

$$\text{minimize}_{q_{\alpha_1}, q_{\alpha_2}} \frac{(N-1)p(1-q_{\alpha_1}p_s)}{[p+(1-p)q_{\alpha_2}p_s][Np+(1-N)pq_{\alpha_1}p_s+(1-p)q_{\alpha_2}p_s]} \quad (20a)$$

$$\text{subject to } \frac{p(N-1)[p(q_{\alpha_1} - q_{\alpha_2}) + q_{\alpha_2}]}{Np - (N-1)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \leq \eta. \quad (20b)$$

It can be shown that the objective function in (20a) decreases with both q_{α_1} and q_{α_2} . This means that the objective function reaches its minimum when q_{α_1} and q_{α_2} take their maximum values. Using (20b), and noting that $0 \leq q_{\alpha_1} \leq 1$, $0 \leq q_{\alpha_2} \leq 1$, and $q_{\alpha_2} \geq q_{\alpha_1}$, the optimal sampling probabilities $q_{\alpha_1}^*$ and $q_{\alpha_2}^*$ can be calculated as follows:

- When $\eta p_s \geq p(N-1)$: The optimal sampling probabilities are $q_{\alpha_1}^* = \min \left\{ 1, \frac{(1-p)(\eta p_s - p(N-1)) + \eta Np}{\eta(N-1)pp_s + p^2(N-1)} \right\}$ and $q_{\alpha_2}^* = 1$.
- When $\eta p_s < p(N-1)$: If $\eta Np \leq (1-p)(p(N-1) - \eta p_s)$, the optimal sampling probabilities are $q_{\alpha_1}^* = 0$ and $q_{\alpha_2}^* = \min \left\{ 1, \frac{\eta Np}{(1-p)(p(N-1) - \eta p_s)} \right\}$. Otherwise, if $\eta Np > (1-p)(p(N-1) - \eta p_s)$, the optimal sampling probabilities are $q_{\alpha_1}^* = \min \left\{ 1, \frac{\eta Np - (1-p)(p(N-1) - \eta p_s)}{\eta(N-1)pp_s + p^2(N-1)} \right\}$ and $q_{\alpha_2}^* = 1$.

We now use (6) to formulate the corresponding optimization problem for the CARS policy as follows:

$$\text{minimize}_{p_{\alpha}^c} \frac{(N-1)(1-p_{\alpha}^c p_s)}{p(N-p_{\alpha}^c p_s)[1+(N-2)p_{\alpha}^c p_s]} \quad (21a)$$

$$\text{subject to } \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] \leq \eta. \quad (21b)$$

For an N -state DTMC information source, the time-averaged sampling action under the CARS policy can be expressed as:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\alpha(t) = 1\} \right] = (N-1)pp_{\alpha}^c. \quad (22)$$

Using (22), (21) can be simplified as:

$$\text{minimize}_{p_{\alpha}^c} \frac{(N-1)(1-p_{\alpha}^c p_s)}{p(N-p_{\alpha}^c p_s)[1+(N-2)p_{\alpha}^c p_s]} \quad (23a)$$

$$\text{subject to } (N-1)pp_{\alpha}^c \leq \eta. \quad (23b)$$

The objective function in (23a) decreases with p_{α}^c . This means that to minimize the objective function, we need to find the maximum value of p_{α}^c that satisfies the constraint in (23b). From this constraint, the maximum feasible sampling probability is $p_{\alpha}^c = \min \left\{ 1, \frac{\eta}{(N-1)p} \right\}$.

B. Maximizing the probability of correct actions

The objective of this optimization problem is to determine the optimal actuation probabilities p_{c_1} and p_{c_2} when the source is sampled using the RS, SARS, and CARS policies. This is achieved by maximizing the probability that the CoAU is zero, while ensuring that the time-averaged actuation action remains below a given threshold. We assume that the parameter $0 < \mu \leq 1$ represents the threshold for the time-averaged actuation action. This optimization problem is formulated as follows:

$$\text{maximize}_{p_{c_1}, p_{c_2}} \mathbb{P}[\Delta(t) = 0] \quad (24a)$$

$$\text{subject to } \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{c(t) = 1\} \right] \leq \mu. \quad (24b)$$

When the source is sampled using the RS policy, the time-averaged actuation action can be expressed as follows:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{c(t) = 1\} \right] = p_{\alpha}^r p_s p_{c_1} + (1-p_{\alpha}^r p_s) p_{c_2}. \quad (25)$$

Using (8) and (25), the optimal actuation probabilities $p_{c_1}^*$ and $p_{c_2}^*$ are obtained as follows:

- When $(1-Np)p_{\alpha}^r p_s \leq (N-2)p$: The optimal actuation probabilities are $p_{c_1}^* = \min \left\{ 1, \frac{\mu}{p_{\alpha}^r p_s} \right\}$ and $p_{c_2}^* = 0$.
- When $(1-Np)p_{\alpha}^r p_s > (N-2)p$: If $\mu \geq p_{\alpha}^r p_s$, the optimal actuation probabilities are $p_{c_1}^* = 1$ and $p_{c_2}^* = \min \left\{ 1, \frac{\mu - p_{\alpha}^r p_s}{1 - p_{\alpha}^r p_s} \right\}$. Otherwise, if $\mu < p_{\alpha}^r p_s$, $p_{c_1}^* = \min \left\{ 1, \frac{\mu}{p_{\alpha}^r p_s} \right\}$ and $p_{c_2}^* = 0$.

When the source is sampled using the SARS policy, the time-averaged actuation action is given by:

$$\begin{aligned} & \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{c(t) = 1\} \right] \\ &= \left(\frac{(N-1)pp_s[p(q_{\alpha_1} - q_{\alpha_2}) + q_{\alpha_2}]}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \right) p_{c_1} \\ &+ \left(\frac{Np + (1-N)p(1+p)q_{\alpha_1}p_s}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \right. \\ &\quad \left. + \frac{(1-p)(1+(1-N)p)q_{\alpha_2}p_s}{Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s} \right) p_{c_2}. \end{aligned} \quad (26)$$

Using (9) and (26), the optimal actuation probabilities $p_{c_1}^*$ and $p_{c_2}^*$ are calculated as follows:

- When $p(N-1)(1-p)q_{\alpha_1}p_s \leq p(N-2) - (1-p)(1+(1-N)p)q_{\alpha_2}p_s$: The optimal actuation probabilities are $p_{c_1}^* = \min \left\{ 1, \frac{\mu f(q_{\alpha_1}, q_{\alpha_2})}{g(q_{\alpha_1}, q_{\alpha_2})} \right\}$ and $p_{c_2}^* = 0$.
- When $p(N-1)(1-p)q_{\alpha_1}p_s > p(N-2) - (1-p)(1+(1-N)p)q_{\alpha_2}p_s$: If $(N-1)pp_s[p(q_{\alpha_1} - q_{\alpha_2}) + q_{\alpha_2}] \leq \mu[Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s]$, the optimal actuation probabilities are $p_{c_1}^* = 1$ and $p_{c_2}^* = \min \left\{ 1, \frac{\mu f(q_{\alpha_1}, q_{\alpha_2}) - g(q_{\alpha_1}, q_{\alpha_2})}{Np + (1-N)p(1+p)q_{\alpha_1}p_s + (1-p)(1+(1-N)p)q_{\alpha_2}p_s} \right\}$. Otherwise, $p_{c_1}^* = \min \left\{ 1, \frac{\mu f(q_{\alpha_1}, q_{\alpha_2})}{g(q_{\alpha_1}, q_{\alpha_2})} \right\}$ and $p_{c_2}^* = 0$.

where $f(q_{\alpha_1}, q_{\alpha_2})$ and $g(q_{\alpha_1}, q_{\alpha_2})$ are given by:

$$\begin{aligned} f(q_{\alpha_1}, q_{\alpha_2}) &= Np + (1-N)pq_{\alpha_1}p_s + (1-p)q_{\alpha_2}p_s, \\ g(q_{\alpha_1}, q_{\alpha_2}) &= (N-1)pp_s[p(q_{\alpha_1} - q_{\alpha_2}) + q_{\alpha_2}]. \end{aligned} \quad (27)$$

When the source is sampled using the CARS policy, the time-averaged actuation action can be expressed as:

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{a(t) = 1\} \right] \\ = [(N-1)pp_{\alpha}^c p_{c_1} + (1 + (1-N)pp_{\alpha}^c p_s)p_{c_2}]. \end{aligned} \quad (28)$$

Using (11) and (28), the optimal actuation probabilities $p_{c_1}^*$ and $p_{c_2}^*$ are obtained as detailed below.

- When $(N - p_{\alpha}^c p_s)[2N - 3 + (N-1)pp_{\alpha}^c p_s] \geq 2(N-1)^2$: The optimal actuation probabilities are $p_{c_2}^* = 0$ and $p_{c_1}^* = \min \left\{ 1, \frac{\mu}{(N-1)pp_{\alpha}^c p_s} \right\}$.
- When $(N - p_{\alpha}^c p_s)[2N - 3 + (N-1)pp_{\alpha}^c p_s] < 2(N-1)^2$: If $(N-1)pp_{\alpha}^c p_s < \mu$, the optimal actuation probabilities are $p_{c_1}^* = 1$ and $p_{c_2}^* = \min \left\{ 1, \frac{\mu - (N-1)pp_{\alpha}^c p_s}{1 + (1-N)pp_{\alpha}^c p_s} \right\}$. Otherwise, if $(N-1)pp_{\alpha}^c p_s \geq \mu$, $p_{c_1}^* = \min \left\{ 1, \frac{\mu}{(N-1)pp_{\alpha}^c p_s} \right\}$ and $p_{c_2}^* = 0$.

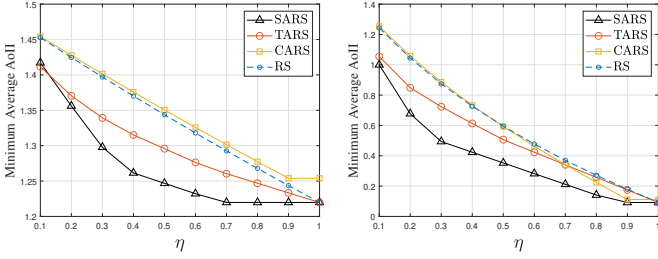
V. NUMERICAL RESULTS

In this section, we numerically evaluate and discuss the analytical results for a $N = 3$ -state Markov chain information source. We assess and compare the average AoII under the optimal RS, CARS, SARS, and TARS policies. We then investigate the probability that the CoAU equals zero under the optimal and non-optimal randomized actuation policies.

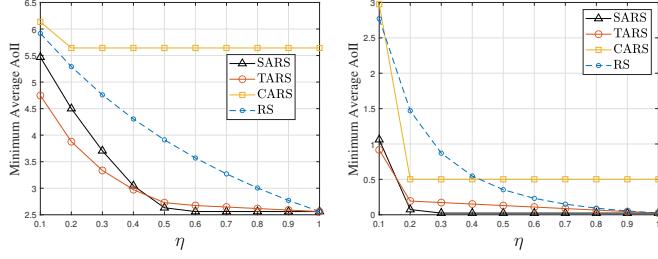
Figs. 2 and 3 show the minimum average AoII under the time-averaged sampling action constraint as a function of η for selected values of q and the success probability p_s . We observe that for a rapidly changing source (i.e., a smaller value of q), where the system must take more sampling actions to reach a synced state, the optimal SARS policy achieves superior performance compared to the other policies. In fact, the main advantage of the optimal SARS policy is that it maintains a low average AoII while avoiding unnecessary sampling actions. Under the SARS policy, sampling at time slot t is performed *probabilistically* only when the system is in an erroneous state, i.e., when $X(t) \neq \hat{X}(t-1)$. *In this policy, more sampling actions are performed when the system remains in an erroneous state for several consecutive time slots than*

when the system was synced in the previous slot but becomes erroneous in the current slot, i.e., $q_{\alpha_2} \geq q_{\alpha_1}$. This adaptive behavior reduces the average AoII and improves overall system performance. For example, Fig. 2(a) illustrates that, for a target average AoII of 1.3, the optimal RS policy reduces the sampling action from 0.70 to 0.67, which corresponds to a 4.2% reduction compared to the optimal CARS policy. The optimal TARS policy further reduces the sampling action to 0.48, achieving a 31.4% reduction relative to the optimal CARS policy. In contrast, the optimal SARS policy requires 57.1% fewer sampling actions (to 0.3) than the optimal CARS policy. *This ability to balance sampling frequency is important for improving performance in dynamic systems and makes the SARS policy suitable for resource-constrained environments such as energy-limited wireless networks.* In contrast, when the source remains in the same state with high probability and the allowed time-averaged sampling action is small (i.e., a larger value of q and a smaller value of η), the optimal TARS policy performs better than the other policies. However, when the sampling action budget is large, the optimal SARS policy achieves better performance. This is because, under a strict sampling action constraint, the optimal SARS policy samples with very low probability even when the system is in an incorrect state. In this case, the TARS policy becomes more effective, as it triggers sampling only when the AoII is large enough to affect system performance significantly.

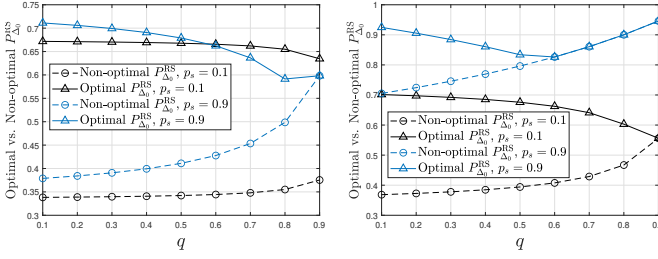
Figs. 4, 5, and 6 illustrate the optimal and non-optimal probability that the CoAU equals zero as a function of q for $\mu = 1$, and for selected values of η and the success probability p_s . In these figures, the non-optimal P_{Δ_0} is obtained by setting $p_{c_1} = p_{c_2} = 1$. For all policies, the sampling and transmission processes use their respective optimal probabilities. The results indicate that when q is small, the probability of taking a correct action decreases. This is because a small q represents a rapidly changing source, which requires more frequent sampling. In this case, the sampling probabilities in the RS, CARS, and SARS policies decrease, increasing the likelihood that the system remains in an erroneous state. By choosing optimal values for p_{c_1} and p_{c_2} , the system can avoid incorrect actions with higher probability, which increases the number of correct actions. For example, when $q = 0.1$ and $\eta = 0.1$, which indicate a rapidly changing source with low sampling and transmission rates, the probability of correct actions increases from 0.34 to 0.67 for $p_s = 0.1$. This corresponds to a 97% increase in correct actions compared to non-optimal actuation. Furthermore, for $p_s = 0.9$, the probability of correct actions increases by 86.8% (from 0.38 to 0.71) under the optimal RS and CARS policies, while under the optimal SARS policy it increases by 72.5% (from 0.40 to 0.69). *It is worth noting that this improvement in correct actions results from both performing correct actions and avoiding incorrect actions when the receiver is uncertain about the correctness of the reconstructed source state.* In addition, increasing η leads to a higher number of correct actions. Since η represents the total time-averaged sampling action, a larger η allows higher sampling and transmission probabilities, which improves the



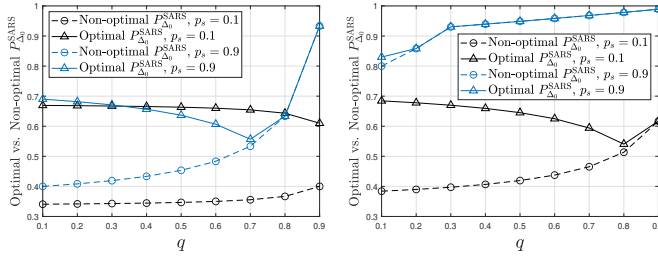
(a) $p_s = 0.1$ (b) $p_s = 0.9$
Fig. 2. Minimum average AoII as a function of η for $q = 0.1$.



(a) $p_s = 0.1$ (b) $p_s = 0.9$
Fig. 3. Minimum average AoII as a function of η for $q = 0.8$.



(a) $\eta = 0.1$ (b) $\eta = 0.7$
Fig. 4. Comparison between the optimal and non-optimal P_{Δ_0} when the source is sampled using the optimal RS policy as a function of q for $\mu = 1$.

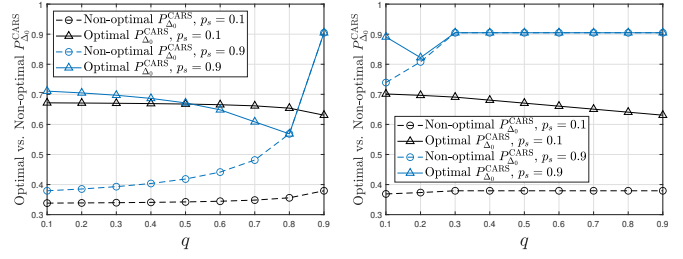


(a) $\eta = 0.1$ (b) $\eta = 0.7$
Fig. 5. Comparison between the optimal and non-optimal P_{Δ_0} when the source is sampled using the optimal SRS policy as a function of q for $\mu = 1$.

accuracy of the system's actions.

VI. CONCLUSION

We studied real-time monitoring of an N -state Markov process in a time-slotted system and derived closed-form expressions for the average AoII under randomized stationary, change-aware randomized stationary, semantics-aware randomized stationary, and threshold-aware randomized stationary sampling and transmission policies. We also proposed a randomized actuation policy and derived the probability of correct actions. Two rate-constrained optimization problems were formulated to jointly optimize sampling and actuation



(a) $\eta = 0.1$ (b) $\eta = 0.7$
Fig. 6. Comparison between the optimal and non-optimal P_{Δ_0} when the source is sampled using the optimal CARS policy as a function of q for $\mu = 1$.

decisions. Our results demonstrate that the semantics-aware randomized stationary policy outperforms the other policies for rapidly changing sources. In contrast, the threshold-aware randomized stationary policy is more effective for slowly changing sources under limited sampling and transmission budgets. Moreover, the proposed randomized actuation policy substantially reduces incorrect actions and enhances overall system performance.

APPENDIX

A. Proof of Lemma 1

Using the DTMC model that describes the evolution of the AoII metric under the TARS policy, the steady-state probabilities π_i , $\forall i \in \{1, 2, \dots\}$, for the TARS policy can be calculated as:

$$\pi_i = \begin{cases} (N-1)p(1-p)^{i-1}\pi_0, & 1 \leq i \leq n-1, \\ (N-1)p(1-p_\alpha^{th}p_s)(1-p)^{i-1}(1-p_s)^{i-n}\pi_0, & i \geq n, \end{cases} \quad (29)$$

where π_0 is given by:

$$\pi_0 = \frac{p + (1-p)p_s}{N(p + (1-p)p_s) - (N-1)(1-p(1-p_\alpha^{th}))p_s(1-p)^{n-1}}. \quad (30)$$

Using (29) and (30), the average AoII under the TARS policy can be obtained as follows:

$$\overline{\text{AoII}} = \sum_{i=1}^{\infty} i\pi_i = \frac{F(p_\alpha^{th}, n)}{G(p_\alpha^{th}, n)}, \quad (31)$$

where $F(p_\alpha^{th}, n)$ and $G(p_\alpha^{th}, n)$ are given by:

$$\begin{aligned} F(p_\alpha^{th}, n) &= (N-1) \left[(1-p)(p + (1-p)p_s)^2 + (1-p)^n p_s \right. \\ &\quad \times \left(3p^2 - 2p - np^2 - p^3 + np^3(1-p_\alpha^{th}) - p^2 p_\alpha^{th} + p^3 p_\alpha^{th} \right. \\ &\quad \left. \left. + (1-p)p_s(1 + (n-2)p - (n-1)p^2(1-p_\alpha^{th})) \right) \right], \\ G(p_\alpha^{th}, n) &= p(p_s + (1-p_s)p) \left[N(1-p)(p_s + (1-p_s)p) \right. \\ &\quad \left. - (N-1)(1-p)^n(1 - (1-p_\alpha^{th})p)p_s \right]. \end{aligned} \quad (32)$$

Similarly, for the RS policy, (29) can be obtained as follows:

$$\pi_i = \begin{cases} \frac{p+(1-p)p_\alpha^r p_s}{Np+(1-Np)p_\alpha^r p_s}, & i = 0, \\ [(1-p)(1-p_\alpha^r p_s)]^{i-1} [(N-1)p(1-p_\alpha^r p_s)] \pi_0, & i \geq 1. \end{cases} \quad (33)$$

Moreover, under the SARS policy, the steady-state probabilities π_i for all $i \geq 0$ are given by:

$$\pi_i = \begin{cases} \frac{p+(1-p)q_{\alpha_2} p_s}{Np+(1-Np)q_{\alpha_1} p_s + (1-p)q_{\alpha_2} p_s}, & i = 0, \\ [(1-p)(1-q_{\alpha_2} p_s)]^{i-1} [(N-1)p(1-q_{\alpha_1} p_s)] \pi_0, & i \geq 1. \end{cases} \quad (34)$$

Furthermore, for the CARS policy, the steady-state probabilities π_i for all $i \geq 0$ can be obtained as follows:

$$\pi_i = \begin{cases} \frac{1+(N-2)p_\alpha^c p_s}{N-p_\alpha^c p_s}, & i = 0, \\ [1-p(1+(N-2)p_\alpha^c p_s)]^{i-1} [(N-1)p(1-p_\alpha^c p_s)] \pi_0, & i \geq 1. \end{cases} \quad (35)$$

B. Proof of Lemma 2

Using (7) and the total probability theorem, the probability $\mathbb{P}[\Delta(t) = 0]$ can be written as follows:

$$\begin{aligned} \mathbb{P}[\Delta(t) = 0] &= \mathbb{P}[c(t) = 1 | X(t) = \hat{X}(t), \alpha(t) = 0] \mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 0] \\ &+ \mathbb{P}[c(t) = 1 | X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 0] \\ &\times \mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 0] \\ &+ \mathbb{P}[c(t) = 1 | X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 1] \\ &\times \mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 1] \\ &+ \mathbb{P}[c(t) = 0 | X(t) \neq \hat{X}(t), \alpha(t) = 0] \mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 0] \\ &+ \mathbb{P}[c(t) = 0 | X(t) \neq \hat{X}(t), \alpha(t) = 1, h(t) = 0] \\ &\times \mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 1, h(t) = 0] \\ &= p_{c_2} (\mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 0] + \mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 0]) \\ &+ (1-p_{c_2}) (\mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 0] + \mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 1, h(t) = 0]) \\ &+ p_{c_1} \mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 1], \end{aligned} \quad (36)$$

When the source is sampled using the RS policy, the probabilities in (36) can be obtained as follows:

$$\mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 0] = \frac{(1-p_\alpha^r)(p+(1-Np)p_\alpha^r p_s)}{p_\alpha^r p_s + Np(1-p_\alpha^r p_s)}, \quad (37)$$

$$\mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 0] = \frac{p_\alpha^r (1-p_s)(p+(1-Np)p_\alpha^r p_s)}{p_\alpha^r p_s + Np(1-p_\alpha^r p_s)}, \quad (38)$$

$$\mathbb{P}[X(t) = \hat{X}(t), \alpha(t) = 1, h(t) = 1] = p_\alpha^r p_s, \quad (39)$$

$$\mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 0] = \frac{(N-1)p(1-p_\alpha^r)}{Np+(1-Np)p_\alpha^r p_s}, \quad (40)$$

$$\mathbb{P}[X(t) \neq \hat{X}(t), \alpha(t) = 1, h(t) = 0] = \frac{(N-1)p p_\alpha^r (1-p_s)}{Np+(1-Np)p_\alpha^r p_s}. \quad (41)$$

Now, using (36) and eqs. (37)–(41), one can obtain the probability $\mathbb{P}[\Delta(t) = 0]$ when the source is sampled using the RS policy.

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